

Cylinders on a Grid: Coarse 3D Layout Control for Video Generation

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Abstract

We ask how little geometry a person has to draw to control both where people stand and where the camera moves in a generated video. Our answer is a ground plane and one cylinder per person. [TODO: two sentences on the data engine and the LoRA; one sentence of results once measured]

1 Introduction

[TODO: Gap: dense controls need a video; camera control ignores people; 2D layout loses the ground.] [TODO: Idea: the minimal hand-authorable 3D proxy — plane + cylinders + camera.] [TODO: Why now: control adapters on open video models; T2V as data source.]

Contributions.

- The plane-and-cylinders control representation and its rendering specification (Sec. 2.1).
- An automatic data engine that turns text-to-video output into aligned (control video, background image, caption, target video) tuples, with no real footage and no manual labels (Sec. 2.2).
- A LoRA on Wan2.2-Fun-Control that follows drawn layouts and camera paths, with an authoring demo (Sec. 2.3, 3).

2 Method

2.1 Control representation

[TODO: plane frame; grid spacing = median subject height; cylinder radius 0.2h, height fixed from the anchor second; 6-color palette by left-to-right order; black background; painter’s ordering; 81 frames at 16 fps. Why each choice.]

2.2 Data engine

[TODO: Framing: the control signal is new, so no dataset pairs it with video; we build the pairs from a text-to-video model; the design question is coverage exactly where the user is free at inference time (place, appearance, action, camera) while keeping the geometry recoverable.]

Captions. [TODO: P1 principle: seeds from a combinatorial grid guarantee coverage; the language model only invents the specifics inside a seed (free-form collapses into clichés, pure templates cap the number of places).] [TODO: P2 axes and guarantees: 50 regions $\times$ 162 ground archetypes = a unique (place, surface) per caption; 118 action families; 8 camera moves; weather / indoor lighting; environment motion incl. still; 3 motion ranges; 6 person-count buckets; crowd and close-approach flags; balanced decks give exact shares.] [TODO: P3 plausibility: per-card eligibility rules (hard floor for ball sports, room for sprints, open sky for throws, skating-only on ice, ≤ 3 people in small venues) learned from an LLM pre-check whose flag rate converged from 37/2000 to 4.] [TODO: P4 writing and checking: writer (40 cards per call, caption + scene + action summary), independent validator, mechanical length / forbidden-word / global near-duplicate checks, repair with the reason attached; 2000 captions, 2000 distinct places, 0 duplicates. Vocabularies and full rules in Appendix A.]

Videos. [TODO: Veo 3.1 Fast, 6 s, 720p, 24 fps, seed = caption id; resampled to 81 frames at 16 fps by dropping every third frame.]

Lifting clips to geometry. [TODO: SAM 3.1 tracking and LaMa background (Fig. 5); agentic ground mask (propose phrase $\rightarrow$ segment $\rightarrow$ judge $\rightarrow$ retry); HunyuanWorld-Mirror cameras/points/normals on 10 keyframes; RANSAC plane; foot-ray $\cap$ plane with head-ray fallback; QC flags.]

2.3 Training

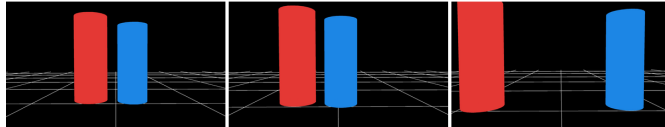
[TODO: Wan2.2-Fun-A14B-Control, control_ref mode (control video + background image + caption); LoRA rank TBD; experts TBD; resolution schedule TBD; N training-ready clips TBD; 40 hold-out.]

2.4 Authoring and inference

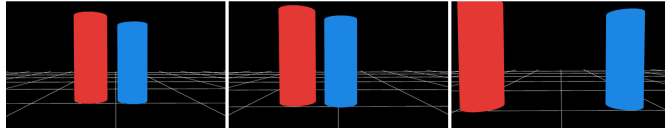
[TODO: minimal editor: place and drag cylinders on the grid over time, move the camera, export the control video; background image + prompt.]

3 Experiments

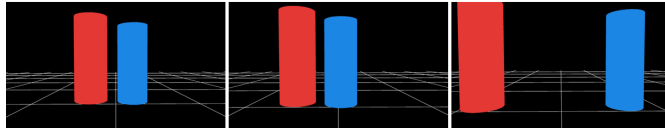
control video (input): ground grid + one cylinder per person, $t = 0\text{ s}, 2.5\text{ s}, 5\text{ s}$
 same geometry · caption A, background A



same geometry · caption B: the two people changed by text (elderly man in tweed, young woman in a yellow raincoat)



same geometry · background C: reference image changed to a frozen lagoon (caption A, scene words adjusted)



generated video



Figure 1: **A ground grid and one cylinder per person are enough to direct a video model.** Left: the control video (three of 81 frames) drawn for two people on a ground plane, seen from a static camera. Right: videos generated from that control video. The rows share the geometry and differ only in the text (row 2) or in the reference image that sets the background (row 3): the people stand where the cylinders stand and keep their left-to-right order and height, while appearance follows the text and the background follows the reference image. [TODO: check final wording]

[TODO: F2 method overview: control representation, data engine, LoRA on Wan2.2-Fun-Control, authoring at inference]

Figure 2: [TODO: overview caption]

[TODO: F3 control spec: render spec diagram + one rendered frame + matching real frame]

Figure 3: [TODO: control spec caption]

Table 1: [TODO: dataset statistics caption]

Item	
Captions / clips	
Distinct (region, ground archetype) pairs	
Action families	
Camera moves	
Person-count buckets (1–6)	500 / 450 / 350 / 300 / 200 /
Training-ready clips after QC	
Hold-out clips	

/ sand; hand-drawn control videos never seen in training.]

Dataset statistics. [TODO: axis coverage; QC pass rates; per-family counts.]

Qualitative results. [TODO: same geometry × prompts; same prompt × geometry; all 8 camera moves; 1–6 subjects with occlusion order; indoor / outdoor / ice

Controllability. [TODO: (a) foot-point reprojection error via SAM 3 on outputs; (b) subject-count accuracy; (c) camera-path consistency via re-reconstruction; (d) reference-free quality score.]

[TODO: F4 data engine: caption -> frame -> person masks -> background -> ground mask -> plane + cylinders -> control frame]

Figure 4: [TODO: data engine caption]



Figure 5: [TODO: Person tracking and background removal. Rows: four pilot clips with one to four subjects, indoor and outdoor, static and moving cameras. Columns: the first frame, SAM 3.1 masks at 0, 2.5 and 5 s (one colour per track, identities held across the clip), and the LaMa background inpainted from the first frame, which becomes the reference image at training time.]

Table 2: [TODO: controllability caption; lower is better for errors]

Method	Foot err. ↓	Count acc. ↑	Cam. err. ↓
Text only	TBD	TBD	TBD
Base + depth render	TBD	TBD	TBD
Base + pose render	TBD	TBD	TBD
Ours	TBD	TBD	TBD

paths; jumps; crossing paths; extreme venues; prompts that contradict the geometry; off-distribution grid spacing or camera height. Each limitation paired with a direction: use → heading / stick figure; single plane → multi-plane; airborne → per-frame height; T2V distribution → real footage with the same engine.]

4 Related Work

Baselines. [TODO: same geometry through depth / pose renders on the base control model; text-only. (Keep only what is cheap.)]

Controllable video diffusion. [TODO: FunControl, VACE, Ctrl-Adapter; pose-guided human video; drag/trajectory control.]

Stress tests and future directions. [TODO: many subjects beyond the training range; long or fast camera

Camera control. [TODO: CameraCtrl, MotionCtrl, ReCamMaster: camera but not people.]

[TODO: F5 qualitative grid: same geometry x different prompts; same prompt x different geometry; 1–6 subjects; indoor / outdoor / ice / sand]

Figure 6: [TODO: qualitative caption]

Layout and 3D-aware control. [TODO: 2D boxes; 3D primitives in image generation; render-then-generate.]

Data engines for control learning. [TODO: adapters trained on generated data; LLM-judged data pipelines.]

Scene understanding tools. [TODO: SAM 3, LaMa, feed-forward multi-view reconstruction (HunyuanWorld-Mirror, VGGT-family).]

5 Conclusion

[TODO: A ground plane and one cylinder per person are enough to tell a video model where people stand, how they move and where the camera goes, while text and a reference image decide how everything looks. The training pairs for such a signal can be built without real footage or manual labels by generating videos and lifting them back to their geometry. Next: richer primitives (heading, stick figure, boxes for vehicles), several planes, real footage lifted by the same engine.]

References

A Caption rules and axis vocabularies

[TODO: from captions/CAPTION_DESIGN.md and axes.json: regions, archetypes, action families with eligibility tags, the 11 rules]

B Agent prompts

[TODO: writer, validator, plausibility pre-check, ground-mask proposer and judge]

Table 3: [TODO: caption axes caption]

Axis	Dealing
Region	unique pair with archetype
Ground archetype	unique pair with region
Action family	least-used-first, eligibility-filtered
Camera move	balanced deck
Weather (outdoor) / lighting (indoor)	balanced deck
Environment motion	14 elements + still (40%)
Motion range	30 / 40 / 30
Person count	500 / 450 / 350 / 300 / 200 / 20
Crowd / close approach	20% / 15% flags

Table 4: [TODO: QC caption; pilot = 10 clips]

Flag	Pilot	Full
ground_fail	0	TBD
plane_fail	0	TBD
count_mismatch	1	TBD
no_anchor	0	TBD
Training-ready	9	TBD

C Preprocessing quality control

[TODO: per-flag counts on the full run]

D Additional results and failure cases

[TODO:]

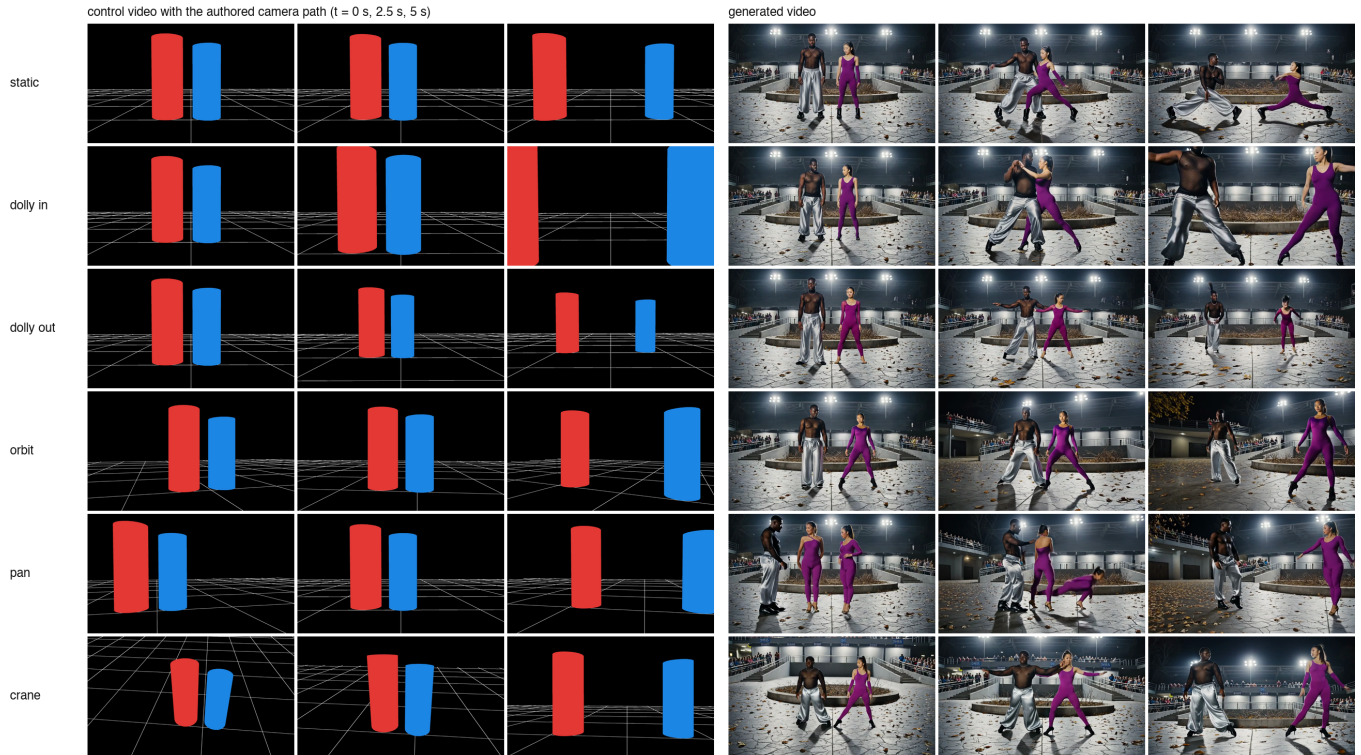


Figure 7: **Camera control.** The same two cylinders, caption and reference image rendered under six authored camera paths (left) and the resulting videos (right). Dolly in, orbit, pan and crane are followed; dolly out is followed only weakly (the subjects shrink less than the cylinders), and the pan clip shows a third person in its first frames. [TODO: add the authored-vs-recovered path plot; check wording]

[TODO: F7 stress tests: one column per test, control frame above generated frame, success and failure marked]

Figure 8: [TODO: stress tests caption]